Technical Supplement to Network G: Reorganized Formal Foundation and Comparative Framework

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Abstract

This reorganized supplement provides the formal field-theoretic defense for the Impedance Phase Persistence (IPP) framework. It establishes a stable, covariant foundation (Pillar 1), extends this structure into a covariant cosmological law (Pillar 2), and defines the non-linear high-stress phase transition regime of the vacuum response (Pillar 3). The purpose of this supplement is formal closure: to place the phenomenological engine of Network G on an explicit augmented Horndeski / covariant EFT backbone that is manifestly stable on the intended branches and internally consistent across galactic, cosmological, and cluster scales.

1 Pillar 1: Formal Foundation & Stability (Closed Covariant Form)

This section replaces previous empirical parameterizations with a formal Lagrangian architecture to address adversarial stability audits.

1.1 Geometric Moment and Coupling Postulate

The theory adopts the exact $SO(3)$-symmetric angular moment of the transverse dipole kernel as its geometric coupling seed:

$$\Omega_{\text{trans}} \equiv \int_{S^2} \sin^2\theta \, d\Omega = \int_0^{2\pi} d\phi \int_0^{\pi} \sin^3\theta \, d\theta = \frac{8\pi}{3}. \tag{1}$$

The corresponding fundamental coupling postulate is:

$$\alpha_0 \equiv \alpha_{\text{fs}} \, \Omega_{\text{trans}} = \alpha_{\text{fs}} \frac{8\pi}{3} \approx 0.06113. \tag{2}$$

Here Ω_{trans} is an exact geometric moment of the transverse kernel on an isotropic sphere. The identification of α_0 with this moment is a physical postulate of the theory, representing the macroscopic gravitational response to vacuum stiffness.

1.2 Covariant Field Content and Explicit Action

To avoid self-projection pathologies and ensure manifest covariance, the theory utilizes two distinct scalar sectors and a smoothing sector. The formal backbone is best described as an *augmented Horndeski / covariant EFT architecture*: the clock and response sectors are organized so that the displayed equations of motion remain second order on the intended branch, while the additional screening and solid sectors are introduced in explicitly first-derivative form.

Define

$$X_\Theta \equiv -\frac{1}{2} g^{\mu\nu}\nabla_\mu\Theta\nabla_\nu\Theta, \qquad X_\varphi \equiv -\frac{1}{2} g^{\mu\nu}\nabla_\mu\varphi\nabla_\nu\varphi, \tag{3}$$

and the preferred clock-flow

$$u_\mu \equiv -\frac{\nabla_\mu\Theta}{\sqrt{2X_\Theta}}, \qquad u^\mu u_\mu = -1, \qquad h_{\mu\nu} \equiv g_{\mu\nu} + u_\mu u_\nu. \tag{4}$$

The matter-density source measured in the clock frame is

$$\rho_m \equiv T_{\mu\nu} u^\mu u^\nu. \tag{5}$$

The total low-stress action is then

$$S = \int d^4x\sqrt{-g}\left[\frac{M_{\rm Pl}^2}{2}R + X_\Theta - V(\Theta) + X_\varphi - \frac{1}{2}m_\varphi^2\varphi^2 - \mathcal{L}_{\rm cl} - \mathcal{L}_{\rm screen}\right] + S_m[\psi_m, \hat{g}_{\mu\nu}], \tag{6}$$

with effective matter metric

$$\hat{g}_{\mu\nu} = A^2(\varphi, \mathcal{C}_{\rm eff})\, g_{\mu\nu} + \frac{B_0}{\Lambda_\varphi^4}\nabla_\mu\varphi\nabla_\nu\varphi + C(\theta)\, u_\mu u_\nu, \qquad \theta \equiv \nabla_\mu u^\mu, \tag{7}$$

and conformal factor

$$A(\varphi, \mathcal{C}_{\rm eff}) = 1 + f(\mathcal{C}_{\rm eff})\frac{\alpha_0}{M_{\rm Pl}}\varphi. \tag{8}$$

The weak-field interaction term used later in Pillar 2,

$$\mathcal{L}_{\rm int} = -\, f(\mathcal{C}_{\rm eff})\frac{\alpha_0}{M_{\rm Pl}}\varphi\, T, \tag{9}$$

is understood as the first-order weak-field reduction of the matter coupling $S_m[\psi_m, \hat{g}_{\mu\nu}]$, not as an independent second coupling.

1.3 The Quartic Closure and Hamiltonian Stability

The action utilizes a positive quartic closure density

$$\mathcal{L}_{\rm cl} \equiv \frac{\beta}{8\Lambda_c^4}\left(h^{\mu\nu}\nabla_\mu\varphi\nabla_\nu\varphi - M_{\rm Pl}^2\mathcal{I}_\theta^2\right)^2, \tag{10}$$

which enters the action as $-\mathcal{L}_{\rm cl}$. In the stiff limit ($\beta \gg 1$), it enforces the algebraic saturation branch

$$h^{\mu\nu}\nabla_\mu\varphi\nabla_\nu\varphi = M_{\rm Pl}^2\mathcal{I}_\theta^2, \tag{11}$$

and hence $|\nabla_\perp\varphi| = M_{\rm Pl}\mathcal{I}_\theta$ on the selected galactic branch.

Around a quasi-static galactic background ($X_\Theta > 0$), the quadratic action for $\delta\varphi$ confirms that the theory is ghost-free and free of Laplacian instability, as the Hamiltonian density is bounded below ($\mathcal{H}^{(2)} \geq 0$) with positive kinetic coefficients:

$$\mathcal{K}_t = 1, \qquad \mathcal{K}_\perp = 1, \qquad \mathcal{K}_\parallel = 1 + \frac{\beta M_{\rm Pl}^2\mathcal{I}_\theta^2}{\Lambda_c^4} > 0. \tag{12}$$

1.4 Compaction Screening and Boundary Recovery

The screening sector is written explicitly through the positive smoothing density

$$\mathcal{L}_{\text{screen}} \equiv \frac{1}{2}\Sigma^2 + \frac{\ell_s^2}{2}\, h^{\mu\nu}\nabla_\mu\Sigma\nabla_\nu\Sigma - \Sigma\,\rho_m, \tag{13}$$

which enters the action as $-\mathcal{L}_{\text{screen}}$. Varying with respect to Σ yields the sourced smoothing equation

$$(1-\ell_s^2\Box_\perp)\Sigma = \rho_m, \qquad \Box_\perp\Sigma \equiv \nabla_\mu(h^{\mu\nu}\nabla_\nu\Sigma)\,. \tag{14}$$

The covariant compaction invariant is then

$$\mathcal{C} \equiv L_*^2\frac{|\Box_\perp\Sigma|}{\Sigma+\Sigma_*}, \tag{15}$$

with explicit screening function

$$f(\mathcal{C}) = \exp\left[-\left(\frac{\mathcal{C}}{\mathcal{C}_*}\right)^2\right]. \tag{16}$$

As $\mathcal{C}\to\infty$, both $f(\mathcal{C})$ and $f'(\mathcal{C})$ vanish. Consequently, the effective matter metric $\hat{g}_{\mu\nu}$ reduces continuously to the GR metric in high-compaction environments (e.g. the Solar System) without a manual branch switch.

Pillar 2: Covariant Cosmological Law

1. Covariant definition of the impedance background. The cosmological impedance is defined as a scalar functional of the accumulated expansion history:

$$\mathcal{S}(x) \equiv 1 + s\,\mathcal{N}(x), \qquad \mathcal{N}(x) \equiv \int_{\tau(x)}^{\tau_0}\frac{\theta}{3}\,d\tau, \qquad \theta \equiv \nabla_\mu u^\mu. \tag{17}$$

Because θ is a scalar and $d\tau$ is proper time along the clock-flow u^μ, $\mathcal{S}(x)$ is manifestly covariant and frame-independent. On a spatially homogeneous FLRW background, this reduces exactly to

$$\mathcal{S}(z) = 1 + s\ln(1+z). \tag{18}$$

2. Minimal Friedmann realization. The background completion treats $\mathcal{S}$ as modulating an effective impedance density:

$$\rho_{\text{imp}}(z) = \rho_{\text{imp},0}\,\mathcal{S}(z) = \rho_{\text{imp},0}\big[1 + s\ln(1+z)\big]. \tag{19}$$

The corresponding Friedmann equation is

$$3M_{\text{Pl}}^2H^2 = \rho_m + \rho_r + \rho_{\text{imp}}, \tag{20}$$

and the resulting effective equation of state is

$$w_{\text{imp}}(z) = -1 + \frac{s}{3\,[1+s\ln(1+z)]} > -1. \tag{21}$$

For $s>0$, the logarithmic law is explicitly non-phantom (quintessence-like), ensuring the global stability of the expansion history and avoiding Big Rip singularities.

3. Coupling to the Pillar 1 screening sector. The cosmological background acts by renormalizing the local compaction trigger:

$$\mathcal{C}_{\rm eff}(x) \equiv \mathcal{C}(x)\,\mathcal{S}(x), \qquad \mathcal{C}_{\rm crit}(z) = \frac{\mathcal{C}_*}{\mathcal{S}(z)}. \tag{22}$$

Thus the same disformal/conformal matter-metric coupling introduced in Pillar 1 feeds directly into local observables through the weak-field reduction

$$\mathcal{L}_{\rm int} = -\,f(\mathcal{C}_{\rm eff})\frac{\alpha_0}{M_{\rm Pl}}\varphi\,T. \tag{23}$$

4. The Genzel Paradox and Threshold Migration. The percent-level background shift at $z \approx 2$ ($s \ln 3 \approx 0.014$) does not act as a linear suppressor but as a **bifurcation trigger**. For galaxies near the compaction boundary ($\mathcal{C} \approx \mathcal{C}_*$), this mild increase in $\mathcal{S}(z)$ is sufficient to push the system across the non-linear Gaussian screening drop. This explains the Genzel trend—where high-redshift massive disks appear more baryon-dominated—as a biased migration of near-critical systems toward the Newtonian branch at earlier epochs.

5. Cosmological Interpretation and the Hubble Tension. The logarithmic evolution establishes a consistent **impedance lag** between early-time (CMB) and late-time observables. By modulating the expansion history $H(z)$ through $\rho_{\rm imp}$, the framework **defines a covariant channel that may contribute to the reconciliation** of the sound horizon at recombination with local distance-ladder measurements, pending a full multi-probe likelihood analysis. This mechanism provides the formal bridge between the $z \approx 1100$ and $z \approx 0$ impedance states without requiring phantom dark energy.

Pillar 3: Vacuum Phase Transitions and the Shatter Wall

1. Covariant Kinetic-Stress Scalar. The non-linear phase transition of the vacuum is governed by a covariant generalization of the kinetic-stress scalar. Let u^μ be the cosmological clock-flow defined in Pillar 1, and let U^μ denote the baryonic bulk-flow four-velocity of a given matter component. Define

$$a^\mu(U) \equiv U^\nu \nabla_\nu U^\mu, \qquad A(U) \equiv \sqrt{h_{\mu\nu}(u)\,a^\mu(U)a^\nu(U)}, \tag{24}$$

and

$$\sigma^2(U,u) \equiv c^2\,h_{\mu\nu}(u)\,(U^\mu - u^\mu)(U^\nu - u^\nu). \tag{25}$$

The stress invariant is then

$$\chi_{\rm s}(x) \equiv \frac{\sigma^2(U,u)}{c^2}\,\frac{A(U)}{a_0}. \tag{26}$$

In the local rest frame, this reduces to the empirical IPP limit

$$\chi_{\rm s} \to \frac{v^2}{c^2}\frac{a}{a_0}, \tag{27}$$

ensuring manifest covariance across all frames.

2. Smooth Phase Order Parameter and the Shatter Wall. The transition from the Viscous Phase (Galactic) to the Solid Phase (Cluster) is encoded by the smooth order parameter

$$\Pi(\chi_{\rm s}) \equiv \frac{1}{2}\left[1+\tanh\left(\frac{\chi_{\rm s}-1}{\Delta_\chi}\right)\right]. \tag{28}$$

The **Shatter Wall** is the continuous crossover centered at $\chi_{\rm s}=1$. Critically, for high-density cluster cores, the condition $\chi_{\rm s} \gtrsim 1$ is typically satisfied **prior to impact**. In this **pre-stiffened state**, the vacuum has already crossed into the solid-like regime, ensuring the gravitational potential is "locked" to the baryonic peaks before hydrodynamic stripping occurs.

3. Relativistic Solid Sector and Tidal Memory. On the high-stress branch, the full action becomes

$$S_{\rm solid} = S - \int d^4x\sqrt{-g}\,\mathcal{L}_{\rm solid}, \tag{29}$$

with positive solid-sector density

$$\mathcal{L}_{\rm solid} = \Pi(\chi_{\rm s})\left[\frac{1}{2}\mu_*(B^{IJ}-\delta^{IJ})(B_{IJ}-\delta_{IJ}) + \frac{1}{2}K_*(B^I{}_I-3)^2\right], \tag{30}$$

where

$$B^{IJ} \equiv h^{\mu\nu}(u)\,\nabla_\mu\Phi^I\nabla_\nu\Phi^J. \tag{31}$$

This yields an effective shear modulus

$$\mu_{\rm eff} = \Pi(\chi_{\rm s})\,\mu_*, \tag{32}$$

allowing the vacuum to sustain the long-lived anisotropic stress $\pi^{ij}_{\rm sol}$ necessary to decouple the gravitational Weyl potential from the local Ricci gas peak.

4. Lensing Decomposition: Ricci Peak versus Weyl Peak. The Bullet Cluster lensing offset is resolved via the covariant split of the optical tidal matrix. Ricci focusing is sourced by the local stress-energy of the shocked X-ray plasma (Fluid Phase), while Weyl focusing, which drives gravitational lensing, is sourced by the tidal field and the anisotropic stress of the **pre-stiffened solid-phase vacuum**:

$$\left(\partial_i\partial_j - \frac{1}{3}\delta_{ij}\nabla^2\right)(\Phi-\Psi) = 8\pi G\,\pi^{ij}_{\rm sol}. \tag{33}$$

In a high-velocity merger, the Weyl peak remains tied to the collisionless galaxy streams because they maintain the structural integrity of the **Solid Phase**, while the Ricci peak remains with the collisional gas. This provides a purely mechanical explanation for the observed lensing displacement as a **sorting of vacuum phases** rather than a collision of dark matter particles.

5. Stability and Cluster Scale-Up. The transition is ghost-free and stable provided the moduli remain positive:

$$\mu_{\rm eff} > 0, \qquad K_{\rm eff} > 0. \tag{34}$$

The longitudinal and transverse sound speeds are then

$$c_L^2 = \frac{K_{\rm eff} + \frac{4}{3}\mu_{\rm eff}}{\rho_{\rm sol}+p_{\rm sol}} > 0, \qquad c_T^2 = \frac{\mu_{\rm eff}}{\rho_{\rm sol}+p_{\rm sol}} > 0. \tag{35}$$

This mechanism addresses the cluster scale-up problem: as tidal strain $\chi_{\rm s}$ rises at cluster scale, the vacuum shear modulus increases, providing the additional non-local support previously attributed to dark halos.

6. Empirical Status Note. While Pillar 3 provides a mathematically closed, covariant, and stable mechanism for the separation of gravitational and baryonic peaks (thereby removing the structural deficit of single-scalar theories), it is presented here as a **formal resolution**. Full empirical replacement of collisionless dark matter paradigms is deferred to future high-resolution relativistic merger simulations required to verify exact lensing convergence maps against this phase-transition Lagrangian.

4: Numerical Implementation: The SPARC Engine

Sections 4–11 of the original supplement are retained as the numerical realization of the Pillars above. The piecewise engine (a_{IPP}) is the discretized limit of the Pillar 1 action for high-aspect-ratio disk geometries. Validation is performed across two tiers: a **Strict Blind Audit** ($N = 54$) for predictive integrity and a **Global Convergence Set** ($N = 149$) for statistical closure.

Table 1: 4.1 Comparative Empirical Benchmarks: Blind Audit vs. Global Convergence.

Metric	**Tier 1: Blind Audit ($N = 54$)**	**Tier 2: Global Set ($N = 149$)**
Pearson Correlation (r)	0.9644	—
Coeff. of Determination (R^2)	0.9082	**0.9563**
Primary Precision Metric	$\chi^2/N = 3.7317$	**RMSE = 18.21 km/s**
Total Data Points	1,084	**3,137**

Table 2: 4.2 Comparative Multi-Scale Trace: Observed Velocity and Impedance Recovery.

Scale / Test	**Network G (Engine)**	**Recovery of Target**	**Precision / Error**
Local (Solar System)	Null Deviation (10^{-18})	100% of GR	10^{-18}
Galactic (SPARC)	$R^2 = 0.9563$	99.8% Convergence	±18.21 km/s
Viscous (SLACS)	5.36 / 4.80 Boost	99.8% of Target	0.17% Error
Bullet Cluster	2.3336 Ratio (Static)	99.98% of Offset	0.016% Dev. (Formal)

Note: The Bullet Cluster recovery refers to the static-limit potential ratio $\Phi_{\mathrm{W}}/\Phi_{\mathrm{R}}$ on the Solid-Phase branch. As noted in Pillar 3, full dynamical verification requires relativistic merger simulations.

4.3 Numerical Stability and Parameter Hygiene

Table 3: Parameter Hygiene: Universal vs. Nuisance Variables.

Category	**Parameters**	**Count**	**Status**
Universal (Global)	α_0, n, γ, s	4	Fixed (All 149 Galaxies)
Stellar Population	$\Upsilon_{3.6}$	1	Floated (Nuisance)
Spatial/External	D, i	2	Fixed (SPARC Standard)